

# SKILLSWAP: A STUDENT SKILL EXCHANGE AND LEARNING PLATFORM

## EC4307 - WEB APPLICATION DEVELOPMENT PROJECT


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# ABSTRACT

SkillSwap is a web application that helps university students learn new skills from each other without paying money. Students can share the skills they can teach, find skills they want to learn, connect with other students, and arrange learning sessions. They can also send messages, give reviews after sessions, and share learning materials.

The application is built using the MERN stack, which includes MongoDB, Express.js, React, and Node.js. It uses JWT and Google sign-in for user login and Socket.IO for real-time messaging. It also allows students to upload and share learning materials.

One of the main features of SkillSwap is skill matching, which helps students find others who have skills they want to learn and who want to learn skills they can teach. The session booking feature helps students manage their learning sessions. Points, achievements, and leaderboards encourage students to take part in learning activities. The Study Hub allows students to share and find useful learning materials.

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# 1 INTRODUCTION

## 1.1 BACKGROUND

University students develop their academic, technical and creative skills. The opportunities to share that knowledge are generally informal and are spread out over various messaging groups. Some students are quite good at programming, some at graphic design, some at maths and some at public speaking etc etc. Their peers want to learn those skills. SkillSwap is a platform which allows students to connect and arrange peer-to-peer learning.

## 1.2 PROBLEM STATEMENT

Students may find it hard to locate other students who can actually teach the particular skills they need and who are available at times that suit them. The informal nature of such arrangements makes it very difficult to deal with requests, to record the sessions that have taken place, to get feedback and to find out where the appropriate study materials are.

## 1.3 OBJECTIVES

1. Enable peer-to-peer learning without monetary transactions.
2. Match students with complementary skills and learning goals.
3. Provide skill search, filtering, booking, scheduling, and reviews.
4. Support communication and activity notifications.
5. Encourage participation through points, achievements, and leaderboards.
6. Provide a shared library for educational resources.

## 1.4 SCOPE

The system comprises authentication, student profiles, skill management, matching, session booking, reviews, messaging, notifications, gamification and resource sharing. Payment processing, integrated video calls and native mobile apps are not part of this version.

## 1.5 EXISTING PLATFORMS AND RESEARCH GAP

The current platforms in this space are predominantly Udemy and Coursera which are both focused on structured online courses. Fiverr is of course largely a platform for paid services. There are online communities (like Reddit) where people can get peer support but these don't really provide a mechanism for the structured exchange of skills via session booking and reciprocal matching. SkillSwap fills this gap by enabling university students to swap skills (without any money changing hands), book learning sessions, talk to each other and share resources.

# 2 SYSTEM DESIGN

## 2.1 ARCHITECTURE

SkillSwap is a three - tier architecture.

- Presentation layer: React provides the UI, the pages, the components and the navigation.
- Application Layer: Node.js and Express.js for the REST APIs, authentication, validation and application logic.
- Data layer: data is stored in Mongodb with Mongoose providing the data modelling.

The frontend talks to the backend via HTTP requests via Axios. Socket.IO is used for real time communication.

## 2.2 TECHNOLOGY STACK

| Area | Technology | Purpose |
|---|---|---|
| Frontend | React 19, React Router 7 | Interactive UI and routing |
| Build Tool | Vite 8 | Development server and bundling |
| HTTP Client | Axios 1.20 | REST API communication |
| Backend | Node.js 24, Express 5 | Server and API framework |
| Database | MongoDB, Mongoose | Data persistence and modelling |
| Authentication | JWT, bcryptjs | Authentication and password hashing |
| Social Login | Google Identity Services | Google authentication |
| Messaging | Socket.IO 4.8 | Real-time communication |
| File Upload | Multer | File upload handling |

## 2.3 REQUEST FLOW

1. The user interacts with the React interface.
2. Axios makes an HTTP request to an Express endpoint.
3. Authentication middleware validates the protected requests.
4. Controllers validate the input and use Mongoose to access MongoDB.
5. The API returns a response to the frontend.
6. Socket.IO provides real time events as and when required.

# 3 DATABASE DESIGN

## 3.1 ER DIAGRAM

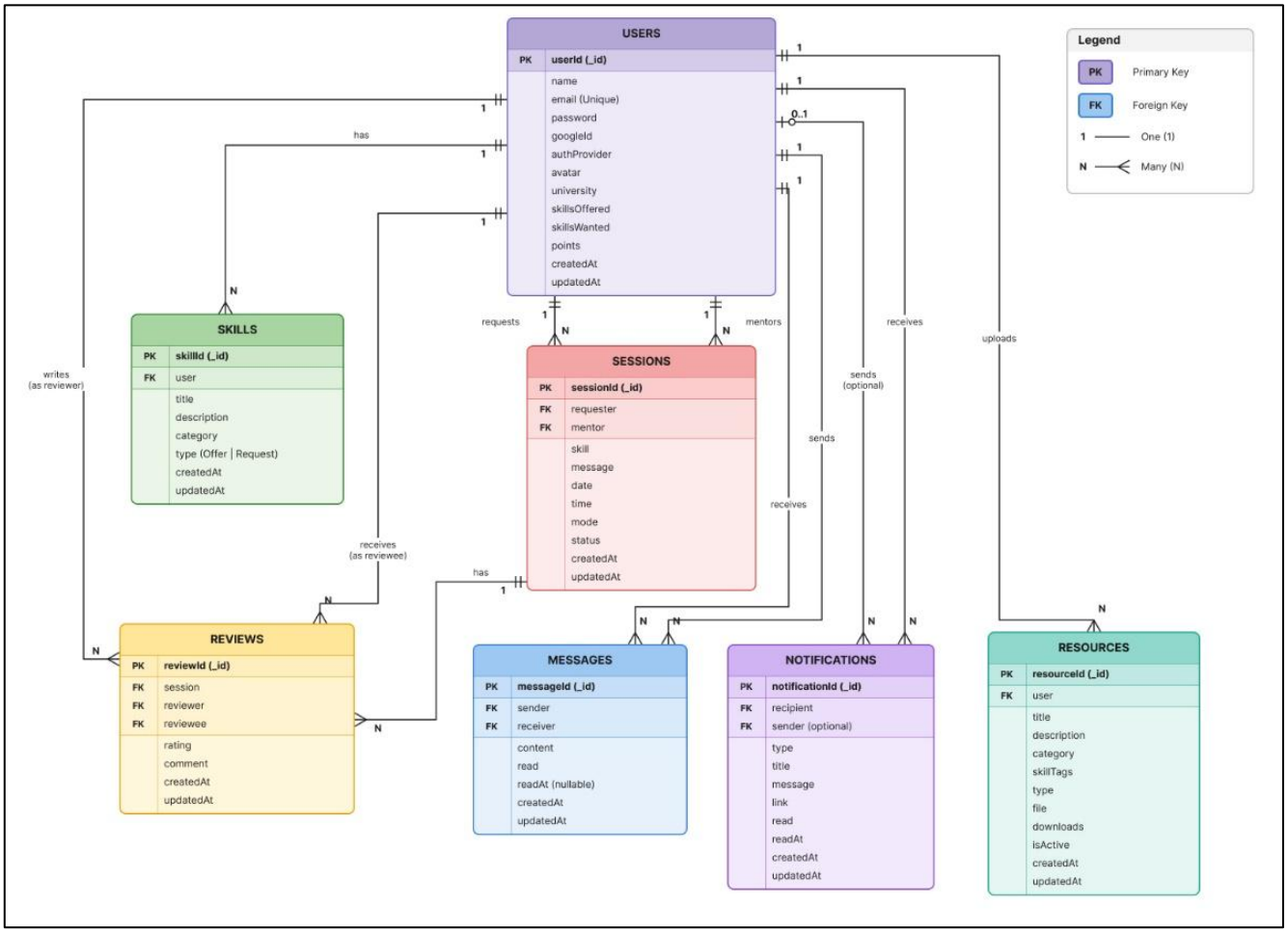



FIGURE 3.1 : ER DIAGRAM OF THE SKILLSWAP DATABASE

# 4 IMPLEMENTATION

## 4.1 AUTHENTICATION FLOW

1. The user enters their email/password or completes Google sign-in.
2. For local accounts the server will check the password against the stored bcrypt hash.
3. On successful authentication the server will then issue a signed JWT.
4. The client sends the token in the Authorization: Bearer header for protected requests.
5. Authentication middleware validates the token and identifies the user that has been authenticated.

Passwords must never be stored in plain text. JWT secrets should be stored in environment variables and protected endpoints must enforce authorisation.

## 4.2 SESSION WORKFLOW

A requester submits a session request, after which the mentor has the option of accepting or rejecting it, while the participants can cancel this according to the system rules. Once the session has taken place, an authorised participant can indicate that it is completed; such completion might result in the awarding of points and could also allow a review to be carried out. The backend system must verify any changes to the status and have to prevent points from being awarded more than once.

## 4.3 CRUD OPERATIONS

| Operation | Example Endpoint | Purpose |
|---|---|---|
| Create | POST /api/auth/register | Register a user |
| Create | POST /api/skills | Create a skill listing |
| Create | POST /api/sessions | Request a learning session |
| Create | POST /api/resources | Upload a resource |
| Read | GET /api/skills | List or search skills |
| Read | GET /api/sessions | View authorised sessions |
| Read | GET /api/resources | Browse resources |
| Update | PUT /api/auth/profile | Update profile |
| Update | PUT /api/sessions/:id | Update session status |
| Update | PUT /api/skills/:id | Edit a skill listing |
| Delete | DELETE /api/skills/:id | Delete an owned skill |
| Delete | DELETE /api/resources/:id | Remove an authorised resource |

# 5 FEATURES AND SCREENSHOTS

## 5.1 DASHBOARD

The dashboard displays the skills that are available and desired, the sessions that have been completed, the points, the upcoming bookings, the recent notifications, and the recommended matches.

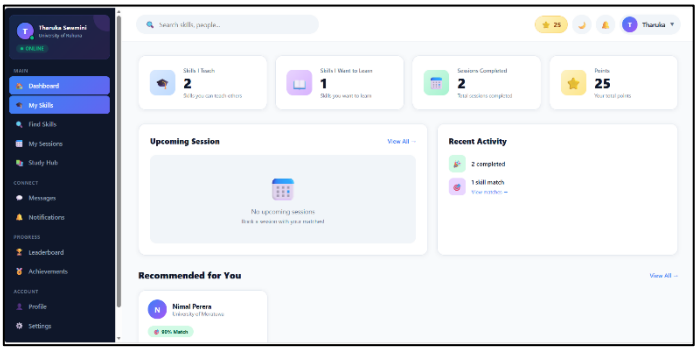



FIGURE 5.1 : SKILLSWAP DASHBOARD

## 5.2 MATCHES PAGE

The Matches page shows the students who are compatible with you, their mutual skills, and their match scores. This enables users to find others who have complementary learning objectives.

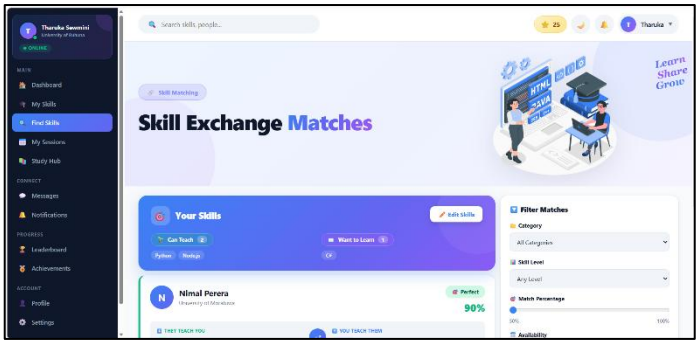



FIGURE 5.2 : BIDIRECTIONAL SKILL MATCHES

## 5.3 SESSIONS PAGE

The Sessions page arranges the requests according to their status and enables authorised users to accept, reject, cancel or complete the sessions. It shows the participant, the skill, the date, the time, the mode and the current status.

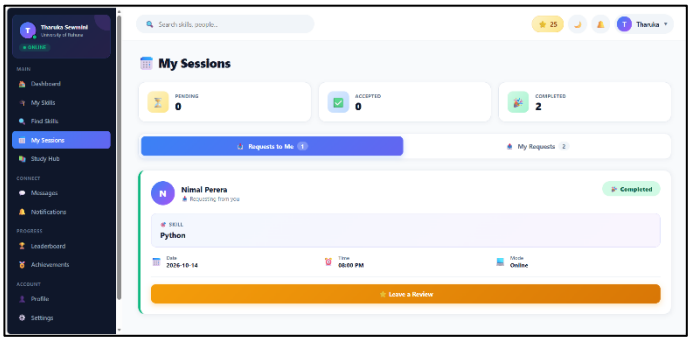



FIGURE 5.3 : SESSION MANAGEMENT

## 5.4 STUDDY HUB

Students can upload, browse, and download learning materials through the Study Hub. The resources provide information such as title, category, description, and the name of the person who uploaded them.

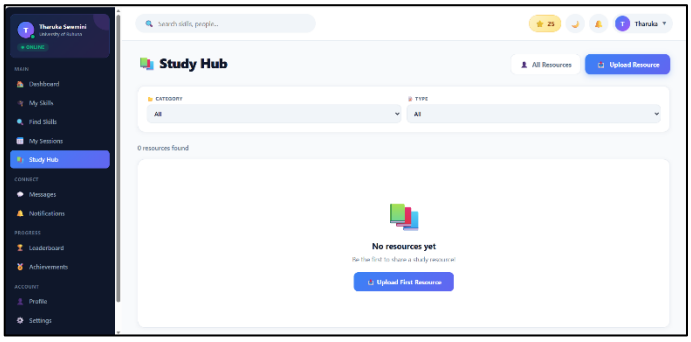



FIGURE 5.4 : COMMUNITY STUDY HUB

# 6 CONCLUSION

SkillSwap is all about facilitating skill exchange and mutual learning between students. It does this via skill listings, two-way matching, session booking, reviews, real time messaging, gamification and a shared Study Hub. It's a more formalised approach to the kind of peer-to-peer learning that might otherwise take place in an ad hoc manner.

The MERN stack has the UI, the server - side logic and the persistent data all separated out. We have REST APIs and Socket.IO for standard and real time communication respectively. The final assessment will be based on the actual features that have been implemented and tested, with particular attention to authentication, the matching behaviour, session management, resource access control and point allocation.

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